

Package ‘lomda’

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Title Longitudinal Omics Multivariate Dimension-reduction Analysis

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Description Implements a two-stage method for the analysis of longitudinal omics data. Stage 1 applies Principal Component Analysis (PCA) to high-dimensional metabolomics (or other omics) measurements collected at multiple visits. Stage 2 fits a Linear Mixed Model (LMM) with a random intercept to the PC scores from Stage 1, enabling inference on time-related trajectories. The package provides estimation, likelihood ratio tests, and Wald tests for the time-effect slope, along with simulated example data and downstream visualization utilities (e.g. PC score plots, loadings, trajectory plots).

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URL <https://github.com/heli-math/lomda>

BugReports <https://github.com/heli-math/lomda/issues>

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lomda	<i>Longitudinal Omics Multivariate Dimension-reduction Analysis (LOMDA)</i>
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Description

Fits the two-stage *lomda* model to longitudinal omics data.

Stage 1: PCA is applied to the omics measurements (metabolites, proteins, etc.) pooled across all visits. The resulting PC scores capture the major axes of variation in the high-dimensional data.

Stage 2: For each selected PC, a Linear Mixed Model (LMM) with subject-specific random intercepts is fitted:

$$t_{ik} = \beta_0 + \beta_1 \cdot k + b_i + g_{ik}$$

where $b_i \sim N(0, \Sigma_b)$ is the random intercept and $g_{ik} \sim N(0, \Sigma_g)$.

Usage

```
lomda(
  data,
  n_pc = 3,
  covariates = NULL,
  scale = TRUE,
  center = TRUE,
  REML = FALSE
)
```

Arguments

<code>data</code>	A data frame with columns: ID (subject identifier), visit (visit; integer 1, 2, 3, ...), age (or other time-invariant covariates), and one column per omics feature. The omics columns must start from column 4 onward.
<code>n_pc</code>	Integer. Number of PCs to extract and model in Stage 2. Defaults to 3.
<code>covariates</code>	Character vector of additional covariate column names to include in the Stage 2 LMM (e.g., <code>c("age", "visit")</code>). Defaults to <code>"visit"</code> .
<code>scale</code>	Logical. Whether to scale the omics features to unit variance before PCA. Defaults to TRUE.
<code>center</code>	Logical. Whether to center the omics features before PCA. Defaults to TRUE.
<code>REML</code>	Logical. Whether to use Restricted Maximum Likelihood (REML) in Stage 2. Defaults to FALSE (ML, required for LRT).

Value

An object of class `"lomda"` with the following components:

`pca` The `prcomp` object from Stage 1.

`scores` Data frame of PC scores merged with metadata (ID, visit, covariates).

`lmm_fits` Named list of `lmerMod` objects, one per PC.

`lmm_null_fits` Named list of null `lmerMod` objects (time dropped) used for LRT.

`n_pc` Number of PCs modelled.

`covariates` Covariate names used in Stage 2.

`call` The matched call.

`data` The input data (metadata + PC scores), stored for downstream plotting.

See Also

[lomda_lrt](#), [lomda_wald](#), [plot.lomda](#), [simulate_lomda_data](#)

Examples

```
dat <- simulate_lomda_data(n_subjects = 50, n_visits = 3, n_features = 20,
  seed = 42)
fit <- lomda(dat, n_pc = 3, covariates = "visit")
print(fit)
summary(fit)
```

lomda_lmm

*Stage 2: LMM on PC scores***Description**

Fits a Linear Mixed Model with random intercept to each PC score derived from Stage 1. This function can be called independently if you already have PC scores.

Usage

```
lomda_lmm(scores_df, n_pc = 3, covariates = "age", REML = FALSE)
```

Arguments

scores_df	A data frame containing at least ID, time, the covariates, and columns named PC1, PC2, etc.
n_pc	Integer. Number of PC columns to model.
covariates	Character vector of additional fixed-effect covariates.
REML	Logical. Use REML? Default FALSE (required for LRT).

Details

The model for the k -th PC is:

$$PC_k = \beta_0 + \beta_1 \cdot \text{time} + \gamma \cdot z + b_i + \varepsilon$$

Value

A list with:

`fits` Named list of lmerMod full-model fits.

`null_fits` Named list of null-model fits (time dropped).

Examples

```
dat <- simulate_lomda_data(seed = 7)
pca_res <- lomda_pca(dat, n_pc = 3)
lmm_res <- lomda_lmm(pca_res$scores, n_pc = 3)
names(lmm_res$fits)
```

lomda_lrt

*Likelihood Ratio Test for the time effect in LOMDA***Description**

Tests whether the time-effect slope (β_1) is significantly different from zero for each PC, by comparing the full LMM (with time) against a nested null model (without time). Both models are fitted by maximum likelihood (REML = FALSE).

Usage

```
lomda_lrt(x)
```

Arguments

x A lomda object (fitted with REML = FALSE).

Details

The LRT statistic is:

$$LRT = -2(\ell_{\text{null}} - \ell_{\text{full}}) \sim \chi^2(1)$$

Value

A data frame with one row per PC and columns:

PC PC name.

logLik_full Log-likelihood of the full model.

logLik_null Log-likelihood of the null model.

LRT_stat Likelihood ratio test statistic.

df Degrees of freedom (1).

p_value P-value from the chi-squared distribution.

Examples

```
dat <- simulate_lomda_data(n_subjects = 60, seed = 99)
fit <- lomda(dat, n_pc = 3, REML = FALSE)
lomda_lrt(fit)
```

lomda_pca

*Stage 1: PCA on longitudinal omics data***Description**

Applies PCA to the omics block of a longitudinal dataset. This is the first stage of the LOMDA pipeline and can be called independently for exploratory purposes.

Usage

```
lomda_pca(data, n_pc = 3, covariates = "age", scale = TRUE, center = TRUE)
```

Arguments

data	A data frame formatted as described in lomda .
n_pc	Integer. Number of principal components to extract.
covariates	Character vector of metadata column names (besides "ID" and "time"). These columns are kept in the returned score data frame. Defaults to "age".
scale	Logical. Scale features to unit variance? Default TRUE.
center	Logical. Center features? Default TRUE.

Value

A list with:

pca The [prcomp](#) object.

scores Data frame: metadata columns + PC score columns.

var_explained Proportion of variance per PC.

loadings Matrix of variable loadings (rotation).

Examples

```
dat <- simulate_lomda_data(n_subjects = 40, n_features = 30, seed = 1)
pca_res <- lomda_pca(dat, n_pc = 5)
head(pca_res$scores)
```

lomda_wald

*Wald Test for the time-effect slope in LOMDA***Description**

Performs a Wald test on the time-effect slope $\hat{\beta}_1$ for each PC. The test statistic is:

$$W = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$$

which is compared against a standard normal distribution (z-test) or, when use_t = TRUE, against a t-distribution with Satterthwaite degrees of freedom (from **lmerTest**).

Usage

```
lomda_wald(x, use_t = TRUE)
```

Arguments

x A lomda object.

use_t Logical. If TRUE (default), use the t-statistic and p-value from **lmerTest**'s Satterthwaite approximation. If FALSE, use a z-test (large-sample approximation).

Value

A data frame with one row per PC and columns:

PC PC name.

beta1 Estimated time-effect slope.

se Standard error of the estimate.

statistic Test statistic (t or z).

df Degrees of freedom (NA for z-test).

p_value Two-sided p-value.

ci_lower 95% confidence interval lower bound.

ci_upper 95% confidence interval upper bound.

Examples

```
dat <- simulate_lomda_data(n_subjects = 60, seed = 99)
fit <- lomda(dat, n_pc = 3)
lomda_wald(fit)
lomda_wald(fit, use_t = FALSE)
```

plot.lomda

Plot method for lomda objects

Description

Convenience wrapper that dispatches to specialized plot functions. The type argument selects which plot to produce.

Usage

```
## S3 method for class 'lomda'
plot(
  x,
  type = "scores",
  pc_x = 1L,
  pc_y = 2L,
  color_by = "visit",
  n_top = 10L,
  ...
)
```

Arguments

x	A lomda object.
type	Character. One of "scores" (PC1 vs PC2 scatter), "loadings" (feature loadings bar chart), "variance" (scree / variance-explained plot), or "trajectory" (mean PC score over visit). Default "scores".
pc_x	Integer. PC for the x-axis in score/loading plots. Default 1.
pc_y	Integer. PC for the y-axis in score/loading plots. Default 2.
color_by	Character. Column name in the score data frame to use for colouring points (e.g., "visit" or "age"). Default "visit".
n_top	Integer. Number of top-loading features to label in the loadings plot. Default 10.
...	Additional arguments passed to underlying plot functions.

Value

A ggplot object (invisibly).

Examples

```
dat <- simulate_lomda_data(seed = 42)
fit <- lomda(dat, n_pc = 3)
plot(fit, type = "scores")
plot(fit, type = "variance")
plot(fit, type = "loadings")
plot(fit, type = "trajectory")
```

plot_loadings	<i>Feature Loadings Plot</i>
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Description

Plots a biplot-style bar chart of feature loadings for one or two PCs, highlighting the top-loading features.

Usage

```
plot_loadings(x, pc_x = 1L, pc_y = 2L, n_top = 10L)
```

Arguments

x	A lomda object.
pc_x	Integer. PC index for x-axis loadings. Default 1.
pc_y	Integer. PC index for y-axis loadings (biplot). If NULL, a simple bar chart of pc_x loadings is produced.
n_top	Integer. Number of features to label / highlight. Default 10.

Value

A ggplot2 object.

Examples

```

dat <- simulate_lomda_data(seed = 5)
fit <- lomda(dat, n_pc = 3)
plot_loadings(fit)
plot_loadings(fit, pc_x = 1, pc_y = NULL)

```

plot_scores	<i>PC Score Plot (PC1 vs PC2 or any two PCs)</i>
-------------	--

Description

Produces a scatter plot of PC scores, coloured by a metadata variable. Each point is one observation (subject x visit).

Usage

```
plot_scores(x, pc_x = 1L, pc_y = 2L, color_by = "visit", label_ids = FALSE)
```

Arguments

x	A lomda object, or a scores data frame (output of lomda_pca).
pc_x	Integer. Index of the PC for the x-axis. Default 1.
pc_y	Integer. Index of the PC for the y-axis. Default 2.
color_by	Character. Column to colour by. Default "visit".
label_ids	Logical. Whether to add subject ID labels with ggrepel . Default FALSE.

Value

A ggplot2 object.

Examples

```

dat <- simulate_lomda_data(seed = 1)
fit <- lomda(dat)
plot_scores(fit)
plot_scores(fit, pc_x = 1, pc_y = 3, color_by = "age")

```

plot_trajectory	<i>Trajectory Plot: Mean PC Score Over Time</i>
-----------------	---

Description

Plots the mean PC score at each visit ($\pm$ standard error), with individual subject trajectories overlaid as thin lines. Useful for visualising the time trend estimated by the Stage 2 LMM.

Usage

```
plot_trajectory(x, pcs = 1:3)
```

Arguments

x	A lomda object.
pcs	Integer vector. Which PCs to plot. Default 1:3.

Value

A ggplot2 object.

Examples

```
dat <- simulate_lomda_data(seed = 10)
fit <- lomda(dat, n_pc = 3)
plot_trajectory(fit)
```

plot_variance_explained	<i>Variance Explained Plot (Scree Plot)</i>
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Description

Displays the proportion (and cumulative proportion) of variance explained by each PC extracted in Stage 1.

Usage

```
plot_variance_explained(x, max_pc = NULL)
```

Arguments

x	A lomda object.
max_pc	Integer. Maximum number of PCs to show. Default: all PCs in the prcomp object (up to 20).

Value

A ggplot2 object.

Examples

```
dat <- simulate_lomda_data(seed = 3)
fit <- lomda(dat, n_pc = 5)
plot_variance_explained(fit)
```

print.lomda	<i>Print method for lomda objects</i>
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Description

Print method for lomda objects

Usage

```
## S3 method for class 'lomda'
print(x, ...)
```

Arguments

x	A lomda object.
...	Ignored.

simulate_lomda_data	<i>Simulate longitudinal omics data for lomda</i>
---------------------	---

Description

Generates a synthetic longitudinal omics dataset suitable for testing and demonstrating the *lomda* two-stage pipeline. The data structure mirrors a typical metabolomics study: multiple subjects measured at several visits, with a latent time trend embedded in the first few features.

Usage

```
simulate_lomda_data(
  n_subjects = 50,
  n_visits = 3,
  n_features = 50,
  n_signal = 5,
  time_effect = 0.5,
  sigma_b = 1.5,
  sigma_e = 1,
  seed = 2026
)
```

Arguments

n_subjects	Integer. Number of subjects. Default 50.
n_visits	Integer. Number of visits per subject. Default 3.
n_features	Integer. Total number of omics features. Default 50.
n_signal	Integer. Number of features with a true time trend. Default 5.
time_effect	Numeric scalar. Slope of the time trend on the signal features. Default 0.5.
sigma_b	Numeric. Standard deviation of subject random intercepts. Default 1.5.
sigma_e	Numeric. Residual standard deviation. Default 1.
seed	Integer. Random seed for reproducibility. Default 2024.

Details

The simulation scheme:

1. Subject-level random intercepts $b_i \sim N(0, \sigma_b^2)$ are drawn for each subject.
2. For each subject i and visit j : $x_{ij,k} = \mu_k + \lambda_k b_i + \beta_k \cdot t_j + \epsilon_{ij,k}$ where λ_k are feature-level loadings (large for the first n_signal features) and β_k is a time trend (non-zero for the first n_signal features).
3. Age is generated as a cross-sectional covariate $\text{age}_i \sim N(45, 10^2)$.

Value

A data frame with columns:

ID Subject identifier (character, e.g. "S001").

visit Visit number (integer 1, 2, ..., n_visits).

age Subject age (numeric).

M1, M2, ... Omics feature values.

The data frame has n_subjects * n_visits rows.

Examples

```
dat <- simulate_lomda_data(n_subjects = 30, n_visits = 3,
                           n_features = 20, seed = 1)
dim(dat)
head(dat[, 1:7])
```

summary.lomda	<i>Summary method for lomda objects</i>
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Description

Prints a table of Stage 2 LMM coefficient estimates (fixed effects) for each PC, including the time-effect slope $\hat{\beta}_1$.

Usage

```
## S3 method for class 'lomda'  
summary(object, ...)
```

Arguments

object	A lomda object.
...	Ignored.

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